FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – VIII
PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**A**	24.	**B**	47.	**A**
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4.	**ABD**	27.	**ABCD**	50.	**ABC**
5.	**BC**	28.	**ABD**	51.	**BD**
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9.	**C**	32.	**C**	55.	**B**
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11.	**4**	34.	**4**	57.	**8**
12.	**1**	35.	**2**	58.	**9**
13.	**2**	36.	**7**	59.	**5**
14.	**6**	37.	**9**	60.	**5**
15.	**4**	38.	**5**	61.	**2**
16.	**3**	39.	**4**	62.	**0**
17.	**5**	40.	**2**	63.	**3**
18.	**2**	41.	**3**	64.	**1**
19.	**1**	42.	**6**	65.	**2**
20.	**4**	43.	**9**	66.	**5**
21.	**00000.73**	44.	**02806.28**	67.	**00102.01**
22.	**00008.89**	45.	**00001.02**	68.	**03600.00**
23.	**00056.56**	46.	**00001.48**	69.	**00007.75**

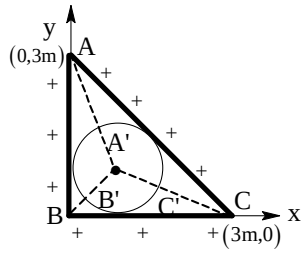
Physics

PART – I

SECTION – A

1. A

Field due to arc A'B' is equal to field due to side AB at the centre of arc. Similarly it will hold for other pairs also. So the field at the incentre is zero.



$$x=\frac{ax_1+bx_2+cx_3}{a+b+c} \Rightarrow y=\frac{ay_1+by_2+cy_3}{a+b+c}$$

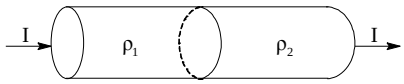
$$x=\frac{3}{2+\sqrt{2}}\,m\ ,\ y=\frac{3}{2+\sqrt{2}}\,m$$

2. C

Since water in the smaller vessel is taking heat from the water in the large vessel. It can reach to a temperature $100°C$. Since the temperature of water in the large vessel cannot go beyond $100°C$ before boiling completely. Hence water in the smaller vessel will not take heat from the water outside, it will not boil.

3. B

Magnitude of flux



$$=\left|E_2-E_1\right|A=\left|\rho_2-\rho_1\right|\ I=\frac{q}{\in_0}$$

$$q=\in_0\ I\left|\rho_1-\rho_2\right|$$

4. ABD

$$T+\frac{2}{3}(200)=20\left(\frac{1}{2}\right)16\Rightarrow T=80/3N \text{ and } \frac{80}{3}+f=10\left(\frac{1}{5}\right)4^2\Rightarrow f=\frac{16}{3}N$$

For maximum value

$$\text{of}\ \omega\Rightarrow T+400/3=20\left(\frac{1}{2}\right)\omega^2\ \&\ T-\frac{2}{3}\times 100=10\left(\frac{1}{5}\right)\omega^2\Rightarrow \omega=5\text{rad/sec}$$

5. BC

$$e=B_0v_0(2y)=4B_0v_0^{3/2}\sqrt{t}$$

$$i=\frac{e}{\lambda(2y)}=\frac{Bv_0}{\lambda} \qquad where\Rightarrow y=\sqrt{4v_0t}$$

$$F=\left(\frac{B_0v_0}{\lambda}\right)(2y)B_0=\frac{2B_0^2v_0y}{\lambda}=\frac{4B_0^2v_0^{3/2}}{\lambda}\sqrt{t}$$

$$p=\vec{F}.\vec{v}=\frac{2B_0^2v_0^2y}{\lambda}=\frac{4B_0^2v_0^{5/2}}{\lambda}\sqrt{t}$$

6. AC

From monkey frame

$$\vec{v}_{am}=\vec{v}_a-\vec{v}_m=40\hat{j}\,m/s$$

$$\text{time}=\frac{40}{40}=1\text{sec}$$

7. BCD

Taking torque about R.H. side of loop

$$\text{mg}\frac{\ell}{2}\text{-T}_1\ell-\text{ibB}=0$$

$$=\text{-}\pi\text{nir}^2\text{B}\hat{\text{k}}$$

8. AD

$$Y=\frac{T/A}{x/\ell} \text{ where } T=M\left(\ell+x\right)\omega^2 \Rightarrow x=\frac{\ell}{\left(\frac{YA}{M\ell\omega^2}-1\right)}$$

And $ms\Delta\theta=\Delta U$

$$\Delta\theta=\frac{YA\ell}{2ms}\left(\frac{1}{\frac{YA}{M\ell\omega^2}-1}\right)^2$$

9. C

10. B

SECTION – C

11. 4

Sol: For solid sphere under pure rolling motion

$$\text{KE}_{\text{translation}}=\frac{5}{7}\text{KE}_{\text{total}}=\frac{1}{2}\text{KX}_{\text{max}}^2$$

$$\therefore \text{X}_{\text{max}}=4\text{cm}$$

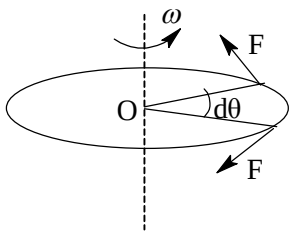
12. 1

$$\frac{\text{V}^2}{\text{R}}=4\left(\text{T-T}_0\right)+4\left(\frac{\text{dT}}{\text{dt}}\right)$$

$$\int_{T_0}^{T}\frac{dT}{\frac{V^2}{R}-4\left(T-T_0\right)}=\int_0^t\frac{dt}{C}\Rightarrow T=T_0+\frac{V^2}{4R}\left(1-e^{-t}\right)\Rightarrow N=1$$

13. 2

Sol.



Hence $dm = \dfrac{1}{2\pi} d\theta$

Hence radial component of tension,

$= 2F \sin \dfrac{d\theta}{2} \simeq F d\theta$

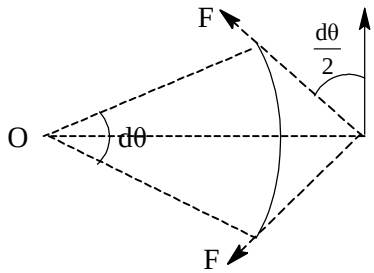
$F d\theta = \dfrac{n d\theta}{2\pi} \omega^2 R \Rightarrow F = \dfrac{m\omega^2 R}{2\pi}$

increment in radius is ΔR so elongation in length of ring

$\sigma = 2\pi\left(R + \Delta R\right) - 2\pi R = 2\pi \Delta R$

Strain $= \dfrac{\sigma}{2\pi R} = \dfrac{\Delta R}{R} \Rightarrow Y = \dfrac{\sigma}{\text{strain}} \Rightarrow \text{strain} = \dfrac{\sigma}{Y}$

$\dfrac{\Delta R}{R} = \dfrac{\sigma}{Y} \Rightarrow \Delta R = \dfrac{m\omega^2 R^2}{2\pi AY} = N = 2$



14. 6

Consider the object as two portion 'A uniform rod' and 'A frustum' with thermal resistance R_1 and R_2 respectively then

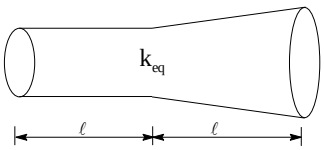
$\mathrm{R}_1 = \dfrac{\ell_1}{k_1 A_1} = \dfrac{\ell}{k\pi r^2}$

And $R_2 = \dfrac{\ell_2}{k_2 A_2} = \dfrac{\ell}{(2k)(\pi r_1 r_2)} = \dfrac{\ell}{4k\pi r^2}$

$\therefore$ Equivalent thermal resistance $R_{eq} = R_1 + R_2$

$\Rightarrow R_{eq} = \dfrac{5\ell}{4k\pi r^2} \ldots\ldots\ldots\ldots\ldots\ldots (1)$

Now if we consider the same lamina with equivalent thermal conductivity K_{eq}, then



$R_{eq} = R_1 + R_2 = \dfrac{\ell}{k_{eq}\pi r^2} + \dfrac{\ell}{k_{eq}\left(2\pi r^2\right)} = \dfrac{3\ell}{2k_{eq}\pi r^2} \ldots\ldots (2)$

By equating the terms of R_{eq} from eqn. (1) & (2), we get

$\dfrac{5\ell}{4k\pi r^2} = \dfrac{3\ell}{2k_{eq}\pi r^2}$

$k_{eq} = \dfrac{6k}{5}$

$\therefore N = 6$

15. 4

As initial angle of approach with horizontal is 45° whereas the river velocity is zero then θ=45°

At point B, $\mathrm{v}_{S/G}$ along the perpendicular to line AB is zero

and $\alpha = 75° - 45° = 30°$

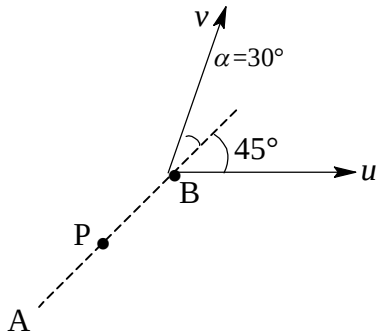
$\therefore v\sin 30° = u\sin 45° \Rightarrow \frac{u}{v} = \frac{1}{\sqrt{2}}$

Also at midpoint P, $v_{river} = \frac{u}{2}$

$\therefore\ v\sin\alpha = \frac{u}{2}\sin\theta \Rightarrow \sin\alpha = \frac{u}{2v}\sin 45° = \frac{1}{4}$

$\alpha = \sin^{-1}\left(\frac{1}{4}\right)$

$\therefore N = 4$

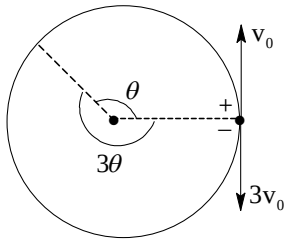


16. 3

$r = \frac{p}{qB} = same\ T_+ = \frac{2\pi\ m_+}{qB} T_- = \frac{2\pi\, m_-}{qB}$

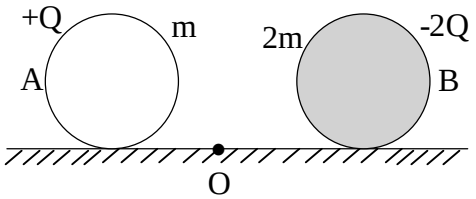
As $T_+ = 3T_-$ they will meet at $\theta = \pi/2$

Time is $\frac{T_+}{4} = \frac{6\pi m}{4qB} = (750\mu s) = 250\times 3\mu s$



17. 5

Let v_1 and v_2 the speed just before collision. Using COAM about point O on the ground



$0 = mv_1R + \frac{2}{3}mR^2\frac{v_1}{R} - 2mv_2R - \frac{2}{5}2mR^2\left(\frac{v_2}{R}\right) \Rightarrow v_1 = \frac{42}{25}v_2$

$K_A = \frac{1}{2}mv_1^2 + \frac{1}{2}\frac{2}{3}mR^2\frac{v_1^2}{R^2} = \frac{1}{2}.mv_1^2\times\frac{5}{3}$

$K_B = \frac{1}{2}2mv_2^2\left(1+\frac{2}{5}\right) = \frac{1}{2}mv_2^2\frac{14}{5}$

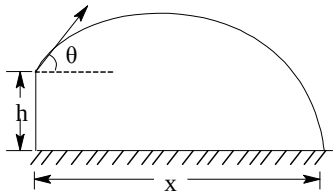
$\frac{K_B}{K_A} = \frac{42}{25}\left(\frac{v_2^2}{v_1^2}\right) = \frac{25}{42}$

18. 2

$x = u_x t - - - - - (1)$

$-h = u_y t - \frac{1}{2}gt^2 - - - - (2)$

Eliminating t and using $\sin\theta = \frac{u_y}{u}$ and $\cos\theta = \frac{u_x}{u}$ and substituting in equation (1) and (2)



$$h\left(1+\cos 2\theta\right) = \frac{gx^2}{u^2} - x\sin 2\theta$$

Differentiating w.r.t.

θ for maximum value of x $\Rightarrow \tan 2\theta = \frac{x}{h} \Rightarrow x = \sqrt{\frac{2u^2}{g}\left(h + \frac{u^2}{2g}\right)}$

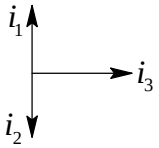
So maximum Area = $\pi x^2 = 209000m^2 \approx 0.2km^2$

19. 1

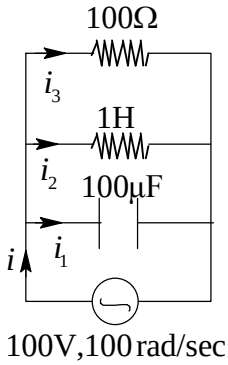
$$\mathrm{X}_L = wL = 100\Omega \Rightarrow X_C = \frac{1}{wc} = 100\Omega$$

$$\mathrm{i}_1 = \frac{100}{100} = 1\mathrm{A} \Rightarrow \mathrm{i}_2 = \frac{100}{100} = 1\mathrm{A} \Rightarrow \mathrm{i}_3 = \frac{100}{100} = 1\mathrm{A}$$

From phasor diagram



So $i = i_3 = 1\mathrm{A}$



20. 4

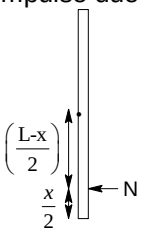
Using WET from a frame moving with velocity $8\left(-\hat{j}\right)m/s$

$$\int_0^{X_{\max}} qv_0B_0x\,dx\cos 180^\circ = -\frac{1}{2}m\left(v_0\sqrt{2}\right)^2 \Rightarrow X_{\max} = 4\,meter$$

SECTION – D

21. 00000.73

Impulse due to 'N' would act at the centre of the colliding parts



Applying Impulse momentum theorem

$$-\int N.dt = M\left(v - v_0\right) \qquad ---(1)$$

& Applying Angular impulse, angular momentum theorem

$$\int N\left(\frac{L-x}{2}\right).dt = \frac{ML^2}{12}\omega \qquad ---(2)$$

Also $e=\dfrac{\omega\left(\dfrac{L-x}{2}\right)-v}{v_0}$ $\quad---(3)$

From (1), (2) & (3)

$$v\left[1+3\left(\frac{L-x}{2}\right)^2\right]=v_0\left[3\frac{(L-x)^2}{L^2}-e\right]$$

$\therefore for\ v>0$

$$x<L\left[1-\sqrt{\frac{e}{3}}\right]$$

For inequality to hold true for all 'e'

$$x<L\left[1-\frac{1}{\sqrt{3}}\right]$$

$$or\ x<\left(\sqrt{3}-1\right),\ \therefore x_{\max}=0.732$$

22. 00008.89

Power absorbed by shell

$$\Rightarrow\frac{\mathrm{P}\times2\pi\left(1-\cos37\right)}{4\pi}$$

$$\Rightarrow\frac{\mathrm{P}}{10}$$

For equilibrium

$$\Rightarrow\frac{P}{10}=3\pi R^2\sigma T^4$$

$$\therefore T^4=\frac{P}{30\pi R^2\sigma}=\frac{2400\sigma\pi}{30\pi\times9\times\sigma}\Rightarrow8.89$$

23. 00056.56

Applying conservation of angular momentum w. r. t. the centre of earth,

$4R\times mv_0\sin30=mvR, v=2v_0$

Applying energy conservation

$$\frac{1}{2}m\left(2v_0\right)^2-\frac{GMm}{R}=-\frac{GMm}{4R}+\frac{1}{2}mv_0^2$$

$$\therefore v_0\Rightarrow\sqrt{\frac{GM}{2R}}\Rightarrow\sqrt{\frac{64\times10^6}{2}}=40\sqrt{2}\times100m/s$$

$$\therefore x=40\sqrt{2}\Rightarrow56.56$$

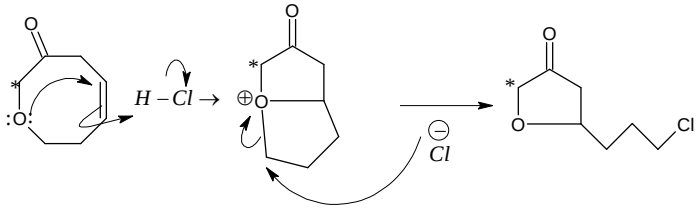
Chemistry

PART – II

SECTION – A

24. B

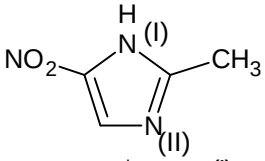
Sol.



25. A

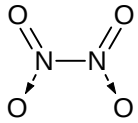
Octanol and water are insoluble.

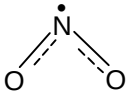
26. D

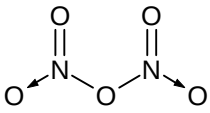
'X' = 

First step is EAS and in 2nd step $N^{(I)}$ is better nucleophile than other $N^{(II)}$.

27. ABCD

 diamagnetic because no unpaired electron

 paramagnetic due to presence of unpaired electron.



HNO_3, H_2SO_4 etc. are responsible for acid rain

28. ABD

$E^0_{Cr^{3+}/Cr^{2+}} < E^0_{Fe^{3+}/Fe^{2+}} < E^0_{Co^{3+}/Co^{2+}}$

29. AD

Hydrolysis requires a $\beta-$galactosidase, showing that galactose and mannose are linked by a $\beta-$galactosidic linkage. For sugar to be reducing, one of the hexoses must have free hemiacetal form.

The methylation /hydrolysis procedure shows the point of attachment of the glycosidic bond to mannose. C_1 is anomeric carbon on mannose ring and both $\alpha-\beta$ forms satisfy the above conditions.

30. ABC

Step-I: $\left[Mg^{2+}\right]=\sqrt[3]{\frac{3.2\times10^{-11}}{4}}=2\times10^{-4}M$

Step-II: $\left[Mg^{2+}\right]=2\times10^{-4}M$

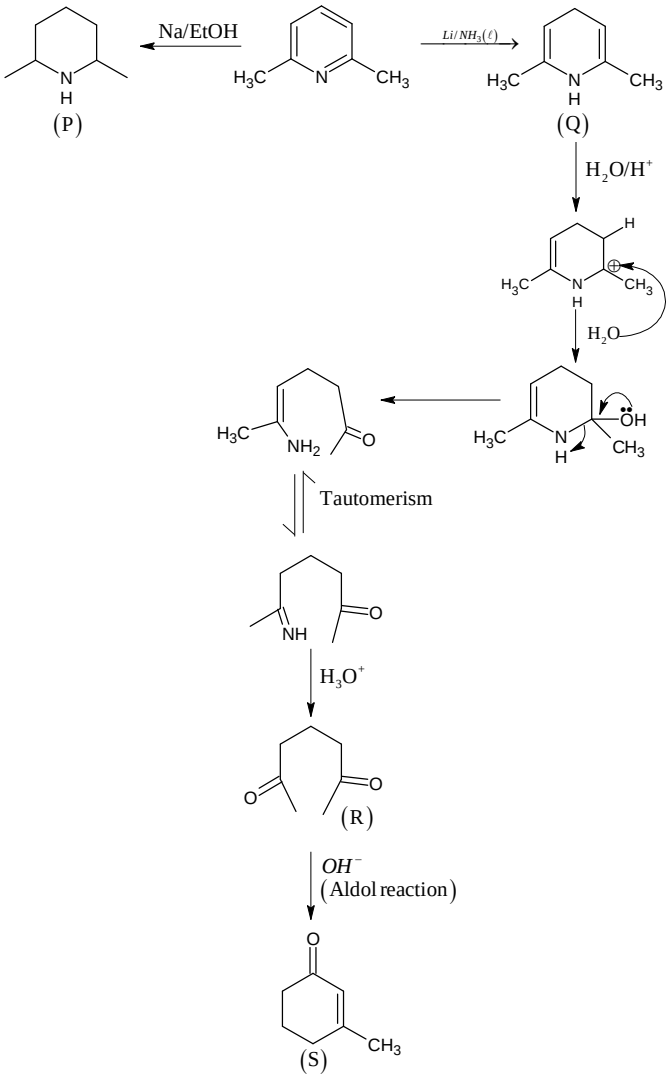
Step-III: $\left[Mg^{2+}\right]=2\times10^{-4}\times\frac{100}{(100+500)}=3.33\times10^{-5}M$

Step-IV:

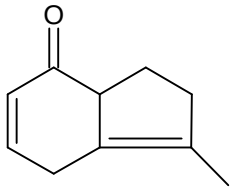
$Mg(OH)_2$ (25ml of 2×10^{-4}M or 0.005 mmole) is mixed with $MgCl_2$ (225ml of 0.06M or 13.5 mmole)

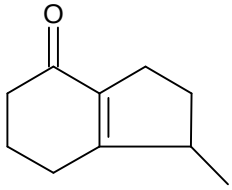
so $\left[Mg^{2+}\right]_{\text{mainly from } MgCl_2}\simeq\frac{13.5}{250}M$

31. ABCD

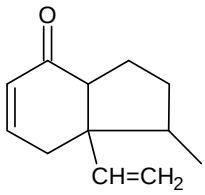


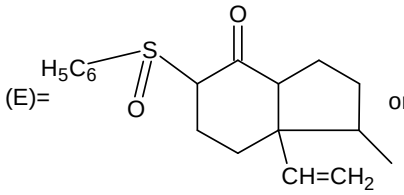
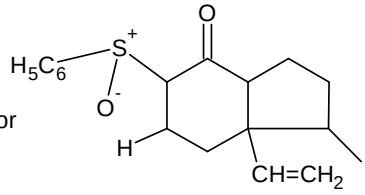
32. C

A= 

B= 

33. A

F= 

(E)=  or 

SECTION – C

34. 4

$K_{a_2} > K_{a_1}$

$\Delta G° = -RT \ln K_a$

$\Delta G_1^\circ > \Delta G_2^\circ$, ΔG_1° and ΔG_2° are both +ve due to $K_a < 1$

$\Delta G_1^0 = \Delta H_1^0 - T\Delta S_1^0$ and $\Delta G_2^0 = \Delta H_2^0 - T\Delta S_2^0$

$\Delta H_1^0 \simeq \Delta H_2^0$

$\Rightarrow \Delta S_1^0$ is more negative than ΔS_2^0.

ΔS_1^0 and ΔS_2^0 both are negative because water is more organised due to hydration.

35. 2

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

$t = 0$ 4 16 –

$t = t_{eq.}$ $4-a$ $16-3a$ $2a$

$P \times 1 = 20 \times R \times T$

$\frac{9}{10} P \times 1 = (20 - 2a) \times R \times T \Rightarrow a = 1$

On adding 'x' moles of inert gas at constant volume, equilibrium composition does not change

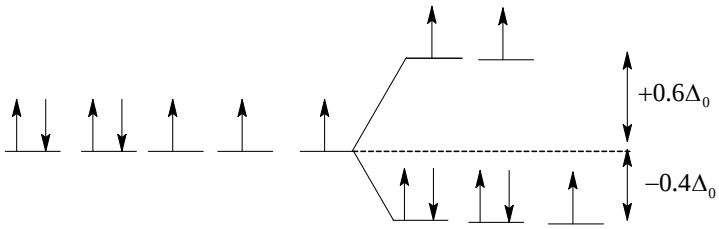
$\frac{9}{10} \times P \times 1 = 18 \times R \times T$

$P \times 1 = (18 + x) RT$

$x = 2$

36. 7

$\text{CFSE} = E_{\text{ligand field}} - E_{\text{symmetric field}}$



$$=\left[5\times\left(\frac{-2}{5}\Delta_0\right)+\left(2\times\frac{3}{5}\Delta_0\right)+2P\right]-2P$$

$$=\frac{-4}{5}\Delta_0$$

$$=\frac{-4}{5}\times 87.5=x=-70$$

$$\Rightarrow\left|\frac{x}{10}\right|=7$$

37. 9

$\Lambda_m=\Lambda_m^0-A\sqrt{C}$

For KCl

$120=150-A\sqrt{0.0625}$

$A=120$

'A' will remains same for KCl and NaCl (1:1 electrolyte)

For NaCl

$$90=126-120\sqrt{\frac{x}{100}}$$

$x=9$

38. 5

$ClF_3, XeF_2, SF_4, [SbF_4]^-$, $XeOF_2$ are having lone pairs at equitorial position.

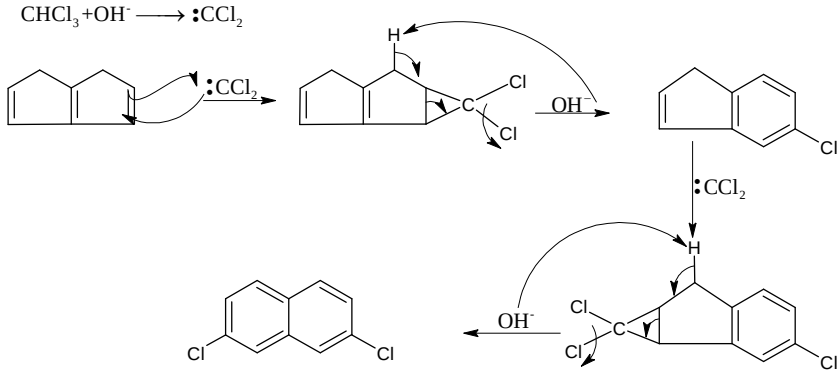
39. 4

degenerate states are





40. 2



It has two plane of symmetry

41. 3

$LiCN, Na_2CO_3, NaHCO_3$

42. 6

Chalcocite Cu_2S, Argentite Ag_2S, Horn silver AgCl, Siderite $FeCO_3$, Azurite $2CuCO_3.Cu(OH)_2$, Limestone $CaCO_3$. $MgCO_3$, Magnesite $MgCO_3$, Kieserite $MgSO_4.H_2O$

43. 9

A and B are same

$$[\alpha]_{sp.rot.} = \frac{\alpha_{obs.}}{c(g/ml)\times l(dm)}$$

$$60 = \frac{\alpha_{obs.}}{\frac{1.5}{10}\times 1}$$

$$\alpha_{obs.} = 9^0$$

SECTION – D

44. 02806.28

$$C_{10}H_8(s) + 12O_2(g) \longrightarrow 10CO_2(g) + 4H_2O(\ell)$$

For 1 mole Napthalene, heat liberated = $\frac{14000\times 2}{0.01} = 28\times 10^5 J/mole$

$\Delta U = -28\times 10^5 J/mol$

$\Delta U_{at\,300K} = -28\times 10^5 - [10\times 2\times 35 + 4\times 75\times 2]$

(but here product and reactant both are at 300K)

$$= (-28\times 10^5 - 1300)J$$

$$= (-2800 - 1.3)kJ = -2801.3kJ$$

$$\Delta H = \Delta U + (\Delta n)_g RT$$

$$= -2801.3 + (-2)8.3\times 300\times 10^{-3}$$

$= -2801.3 - 4.98$

$= -2806.28$ KJ/mol

45. 00001.02

The rate of change of pH with the volume of alkali added is maximum at equivalence point. So volume required to neutralisation = 25.5ml

$25.5 \times 2 \times 0.2 = M \times 1 \times 10$

So, $M = 1.02$

46. 00001.48

$Z = 4 \times 0.999$

$$d = \frac{4 \times 0.999 \times 40}{6 \times 10^{23}\left[(0.18)^{1/3} \times 10^{-7}\right]^3} \text{ gm/cm}^3$$

$= 1.48 \text{ g/cm}^3$

Mathematics

PART – III

SECTION – A

47. A

$2\tan^{-1}\left(\sec^2 \pi x\right) \geq \frac{\pi}{2}$ & $\sin^{-1}\left(x^3 - x^2 + x + 2\right) \leq \frac{\pi}{2}$

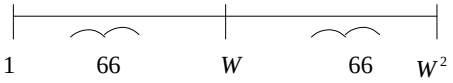
Equality holds if $x^3 - x^2 + x + 2 = 1$ & $\sec^2 \pi x = 1$

$\Rightarrow x^3 - x^2 + x + 1 = 0, \quad x = n, n \in N$ but equation has no integral roots.

48. D

Sol. Factor of $W = 2^4.3^2.5^1 = 5 \times 3 \times 2 = 30$

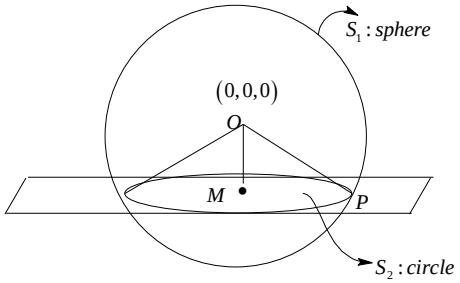
Factor of $W^2 = 2^8.3^4.5^2 = 9 \times 5 \times 3 = 135$



Hence $66 - 28 = 38$

49. A

Sol.



Plane: $x + y + z = 1$

Let OM be $\perp$ distance from 'O' to plane

$\Rightarrow OM = \frac{1}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{1}{\sqrt{3}}$

$OP = 1(given)$

ΔOMP is a right angled triangle

$\therefore MP = \sqrt{1 - \frac{1}{3}} = \sqrt{\frac{2}{3}}$

The required curve is the cone with S_2 as circular base and vertex at $O(0,0,0)$

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times \left(\sqrt{\frac{2}{3}}\right)^2 \times \frac{1}{\sqrt{3}}$

$= \frac{2}{9\sqrt{3}}\pi = \frac{2\sqrt{3}}{27}\pi$

50. ABC

Sol.

$$\left[x^2-1\right]=\begin{cases}0 & x=-1\\ -1 & -1<x<1\\ 0 & 1\le x<\sqrt{2}\\ 1 & \sqrt{2}\le x<\sqrt{3}\\ 2 & \sqrt{3}\le x<2\\ 3 & x=2\end{cases}$$

$$|x|\left[x^2-1\right]=\begin{cases}0 & x=-1\\ x & -1<x<0\\ -x & 0\le x<1\\ 0 & 1\le x<\sqrt{2}\\ x & \sqrt{2}\le x<\sqrt{3}\\ 2x & \sqrt{3}\le x<2\\ 3x & x=2\end{cases}$$

$$|x|+\left[x^2-1\right]=\begin{cases}-x & x=-1\\ -x-1 & -1<x<0\\ x-1 & 0\le x<1\\ x & 1\le x<\sqrt{2}\\ x+1 & \sqrt{2}\le x<\sqrt{3}\\ x+2 & \sqrt{3}\le x<2\\ x+3 & x=2\end{cases}$$

$$\int_{-1/2}^{1/2}\left(f(x)-g(x)\right)dx=\int_{-1/2}^{0}(2x+1)\,dx+\int_{0}^{1/2}(-2x+1)\,dx=\frac{1}{2}$$

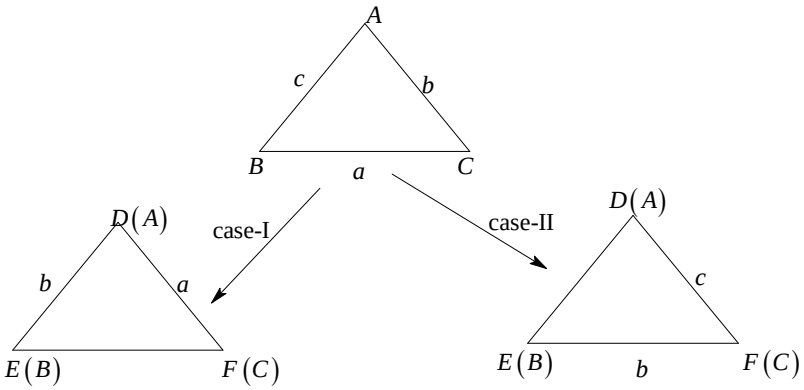
51. BD

Sol. Divide S in 7 partition of form $7k+r\left(r=0,.....,6\right)$ we can choose a element from partition of form $7k+r$. We cannot use $7(k+1)-r.\{r=1,.....,6\}$ and we can select 1 element from partition of form $7k$.

52. ABC

Let $Q(x)=x^2+bx+c, Q\left(P(x)\right)=\left(P(x)\right)^2+bP(x)+c$ as $P(x)\,|\,Q\left(P(x)\right)\Rightarrow c=0$ and $Q(x)=P(x)+b$

53. ACD



case-I

Let $a > b > c$, $\dfrac{\sin C}{\sin B} = \dfrac{c}{b} = \dfrac{b}{a} \Rightarrow b^2 = ac$

case-II

Let $a > b > c$, $\dfrac{\sin A}{\sin B} = \dfrac{b}{c} = \dfrac{a}{b} \Rightarrow b^2 = ac$

Let $(a,b,c) \equiv (c\lambda^2, c\lambda, c)$ as triangle is

formed $b + c > a \Rightarrow \lambda^2 - \lambda - 1 < 0 \Rightarrow \dfrac{2}{1+\sqrt{5}} < \lambda < \dfrac{1+\sqrt{5}}{2}$

54. ABCD

$$\int_0^{\pi} 16\sin^4 x \cos^2 x - \left(f(x) - 4\sin^2 x \cos x \right)^2 dx$$

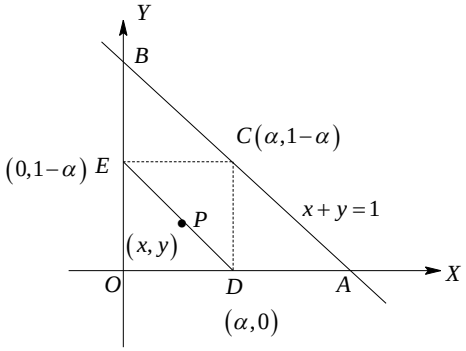
$$= \pi - \int_0^{\pi} \left(f(x) - 4\sin^2 x \cos x \right)^2 = \pi$$

$$f(x) = 4\sin^2 x \cos x$$

55. B

56. D

Sol. (55 – 56)



$\frac{x}{\alpha}+\frac{y}{1-\alpha}=1$

$AM \geq GM$

$\frac{\frac{x}{\alpha}+\frac{y}{1-\alpha}}{2} \geq \sqrt{\frac{xy}{\alpha(1-\alpha)}}$

$\Rightarrow xy \leq \frac{\alpha(1-\alpha)}{4}$

$xy \leq \frac{1}{16}$

Region traced by point P is $(y-x-1)^2 \geq 4x$

$\therefore Area = \int_0^1 \left(x+1-2\sqrt{x}\right)dx = \frac{1}{6}$

SECTION – C

57. 8

Sol. $f(\alpha_i)=\tan\alpha_i \quad e^{-\alpha_i}\left(f(\alpha_i)+\frac{1}{f(\alpha_i)}\right)=\frac{2e^{-\alpha_i}}{\sin 2\alpha_i}=2$

$\sum_{i=1}^{4} e^{-\alpha_i}\left(f(\alpha_i)+\frac{1}{f(\alpha_i)}\right)=8$

58. 9

Sol. $(2\lambda+1,\lambda,-\lambda)$ is the ordered triplet which satisfy first two equations so it must satisfy third equation & $(2\lambda+1)^2+\lambda^2+\lambda^2$ must be minimum which happens at $\lambda=-\frac{1}{3}$ so $(\alpha,\beta,\gamma)\equiv\left(\frac{1}{3},-\frac{1}{3},\frac{1}{3}\right)$ which satisfies third equation, i.e.

$a\times\frac{1}{3}+(a+b)\left(-\frac{1}{3}\right)+6\times\frac{1}{3}=1$

$b=3 \,\&\, a\in R$

59. 5

Sol. Coeff. of x^{14} in $(1-x)^{17}\left(1+2x+3x^2+.....+15x^{14}+.....\right)$ is $\sum_{k=0}^{14}(-1)^k(15-k)\ {}^{17}C_k$.

Which is same as coeff. of x^{14} in $(1-x)^{15}$ $={}^{15}C_{14}=15$

60. 5

Sol. Let $r^{th},(r+2)^{th},(r+6)^{th}$ terms be $a,a+2d,a+6d$ respectively.

$\Rightarrow (a+2d)^2=a(a+6d)\Rightarrow a=2d.$

Terms are $(2d, 4d, 8d)$ so common ratio of G.P is $\frac{1}{2}$.

Let $u, \frac{u}{2}, \frac{u}{2^2}, \ldots\ldots \frac{u}{2^{2p}}$ are $2p+1$ terms in $(p, 2018p)$

i.e.

$u < 2018p \;\&\; \frac{u}{2^{2p}} > p$

$p.2^{2p} < u < 2018p$

$4^p < 2018$

$p \le 5$

61. 2

Sol. $f(x) > x^2 \;\; \forall x \in R - \{\pm 1\}$

$f(f(x)) > f^2(x) \;\; \forall x \in R - \{\pm 1, 0\}$

$f(f(x)) > f^2(x) > (x^2)^2 \quad \forall x \in R - \{\pm 1, 0\}$

$f(f(f(x))) > f^2(f(x)) > f^4(x) > x^8 \quad \forall x \in R - \{\pm 1\}$

$f(f(f(x))) \le x^8$ has solution only at $x = \pm 1$

62. 0

Sol. $\min(f(t)) = -\left(\frac{\sin^2 x - 4 \cdot \frac{1}{4}|\sin x| \cdot 4}{4 \cdot \frac{1}{4}|\sin x|}\right)$

$= 4 - |\sin x|$

$\therefore I_1 = \int_0^{\pi/2} (4 - |\sin x|)\, dx = [4x + \cos x]_0^{\pi/2}$

$I_1 = 2\pi - 1$

$I_2 = \min\left(\int_0^{\pi/2} \left(\frac{1}{4}t^2 |\sin x| - t \sin x + 4\right) dx\right)$

$= \min\left(\left[-\frac{1}{4}t^2 \cos x + t \cos x + 4x\right]_0^{\pi/2}\right)$

$= \min\left(2\pi + \frac{1}{4}t^2 - t\right)$

$= \min\left(2\pi + \frac{1}{4}(t-2)^2 - 1\right)$

$I_2 = 2\pi - 1$

$\therefore |I_1 - I_2| = 0$

63. 3

$(x+2)(2x-3)=p^n$

Case 1: One of the factors is ± 1 which is possible only when $x=2,-3$

Case 2: Let $x+2=p^r$ & $2x-3=p^s$

$\Rightarrow p^s\left(2p^{r-s}-1\right)=7$

$\Rightarrow p=7 \& x=5$

64. 1

Sol. $a^2-bc=b^2-ca=c^2-ab=-(ab+bc+ca)$

So given expression reduces to

$$\frac{(ab)^2+(bc)^2+(ca)^2}{(ab+bc+ca)^2}=\frac{(ab+bc+ca)^2-2abc(a+b+c)}{(ab+bc+ca)^2}$$

$=1$

65. 2

$$\frac{\cos\pi(x+1)}{k}+\frac{\cos\pi(x-1)}{k}=\sqrt{3}\frac{\cos\pi(x)}{k}$$

$$\Rightarrow \frac{\cos\pi(x)}{k}\left(2\frac{\cos\pi}{k}-\sqrt{3}\right)=0$$

$$\Rightarrow \frac{\pi}{k}=\frac{\pi}{6}\Rightarrow k=6$$

$$\lambda\cos\frac{\pi}{3}=4\Rightarrow\lambda=8$$

66. 5

Let chosen points be $(\alpha_1,\beta_1)\&(\alpha_2,\beta_2)$. The area of the triangle $\frac{1}{2}\left|\alpha_1\beta_2-\alpha_2\beta_1\right|$ must be integer.

$\Rightarrow \alpha_1\beta_2-\alpha_2\beta_1$ is even integer. Total probability $=\left(1-\frac{2(64-16).16}{64\times 64}\right)=\frac{5}{8}$

SECTION – D

67. 00102.01

$$\frac{dv}{dt}=-\mu A\Rightarrow\frac{Adh}{dt}=-\mu A$$

$$\int_{15.7}^{0} dh=\int_{0}^{t}-\mu dt \qquad \Rightarrow 15.7=\mu t$$

$$\frac{dv}{dt}=-\lambda \Rightarrow (10.1)^2\frac{dh}{dt}=-\lambda$$

$$\int_{15.7}^{0}(10.1)^2\,dh=\int_{0}^{t}-\lambda dt$$

$$\lambda t=(10.1)^2(15.7)$$

As t is same in both case $\frac{(10.1)^2(15.7)}{\lambda}=\frac{(15.7)}{\mu}\Rightarrow\frac{\lambda}{\mu}=(10.1)^2$

68. 03600.00

As $z_1+z_2+z_3=0\Rightarrow z_1,z_2,z_3$ are vertices of an equilateral triangle and z_4,z_5 are end points of diameter. So $|z_i-z_j|^2=3r^2$ if $i,j\in\{1,2,3\}$; $|z_i-z_j|^2=4r^2$ if $i\in\{1,2,3\}$ & $j\in\{4,5\}$ & $|z_4-z_5|^2=4r^2$

$$\Rightarrow\sum_{1\le i<j\le 5}\sum|z_i-z_j|^2=25r^2=3600$$

69. 00007.75

Let A denotes the probability that every box have three balls each. B is the probability that each box has balls of the colour as that of the box.

Required probability $=P(\bar{B}/A)=1-P(B/A)$

Cases: $R\to B,B\to G,G\to R$ or $R\to G,B\to R,G\to B$

$$P(A)=\frac{1}{2}.\frac{1}{2}.\frac{1}{2}+\frac{1}{2}.\frac{1}{2}.\frac{1}{2}$$

$$P(A\cap B)=\frac{1}{2}.\left(\frac{1}{2}.\frac{1}{4}\right).\left(\frac{1}{2}.\frac{1}{4}\right)$$

So $$P(B/A)=\frac{\frac{1}{8}.\frac{1}{16}}{\frac{1}{8}+\frac{1}{8}}=\frac{1}{32}$$

$$P(\bar{B}/A)=\frac{31}{32}$$